

Tulasimohan Molli

Personal info

Name: Tulasimohan Molli

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I am a PhD candidate in the School of Technology and Computer Science at Tata Institute of Fundamental Research, Mumbai. I am advised by Prof. Prahladh Harsha.

Research Interests

Circuit complexity, Query complexity, Communication complexity, Analysis of Boolean Functions, Coding theory and other related areas in Complexity.

Education

Tata Institute of Fundamental Research 2014- present (in progress)

PhD in Computer Science

Advisor: Prof. Prahladh Harsha

Chennai Mathematical Institute 2012 - 2014

Master of Science in Computer Science

Chennai Mathematical Institute 2009 - 2012

Bachelors of Science(Honours) in Mathematics and Computer Science

Publications

Criticality of AC^0 -Formulae

joint work with Prahladh Harsha and Ashutosh Shankar

- Manuscript under submission to STOC.

Tight Chang's-lemma-type bounds for Boolean functions

joint work with Sourav Chakraborty, Manaswi Parashar, Nikhil Mande, Rajat Mittal and Swagato Sanyal

- In Proc. 41st IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science, FSTTCS 2021, December 15-17, 2021, Virtual Conference
- arXiv:012.02335

On the Probabilistic degree of OR over Reals

joint work with Siddharth Bhandari, Prahladh Harsha and Srikanth Srinivasan

- In *Proc. 38th IARCS Conf. on Foundations of Software Technology & Theoretical Computer Science (FSTTCS)* (Ahmedabad, India, 10-14 December), volume 122 of Leibniz International Proceedings in Informatics, pages 5:1-5:12, 2018. Schloss Dagstuhl.
- *Random Structures and Algorithms 2021*. `rsa.20991`
- `arXiv:1812.01982`

Teaching

Undergrad level course on Discrete Structures and Computation at IISER, Berhampur in Fall of 2022.

Projects and Theses

On Lowerbounds for Depth 2 circuits with threshold and MOD gates

under the guidance of Prof Arkadev Chattopadhyay, TIFR, Mumbai.

Masters Thesis on structural results in Arithmetic circuit complexity and applications to lowerbounds

under the guidance of Prof. Srikanth Srinivasan, IIT Bombay.

Randomized Query Complexity of Recursive Majority

with Prahladh Harsha

Weight distribution and Capacity achievability of Reed-Muller codes in Binary Erasure Channel

with Siddharth Bhandari, Prahladh Harsha, Ramprasad Saptharishi

Criticality and Average case lower bounds for Boolean Formulas

with Yuval Filmus and Prahladh Harsha

Professional work

Reviewed papers for Symposium of STOC, Computational Complexity Conference(CCC) and Symposium on Theoretical Aspects of Computer Science(STACS).